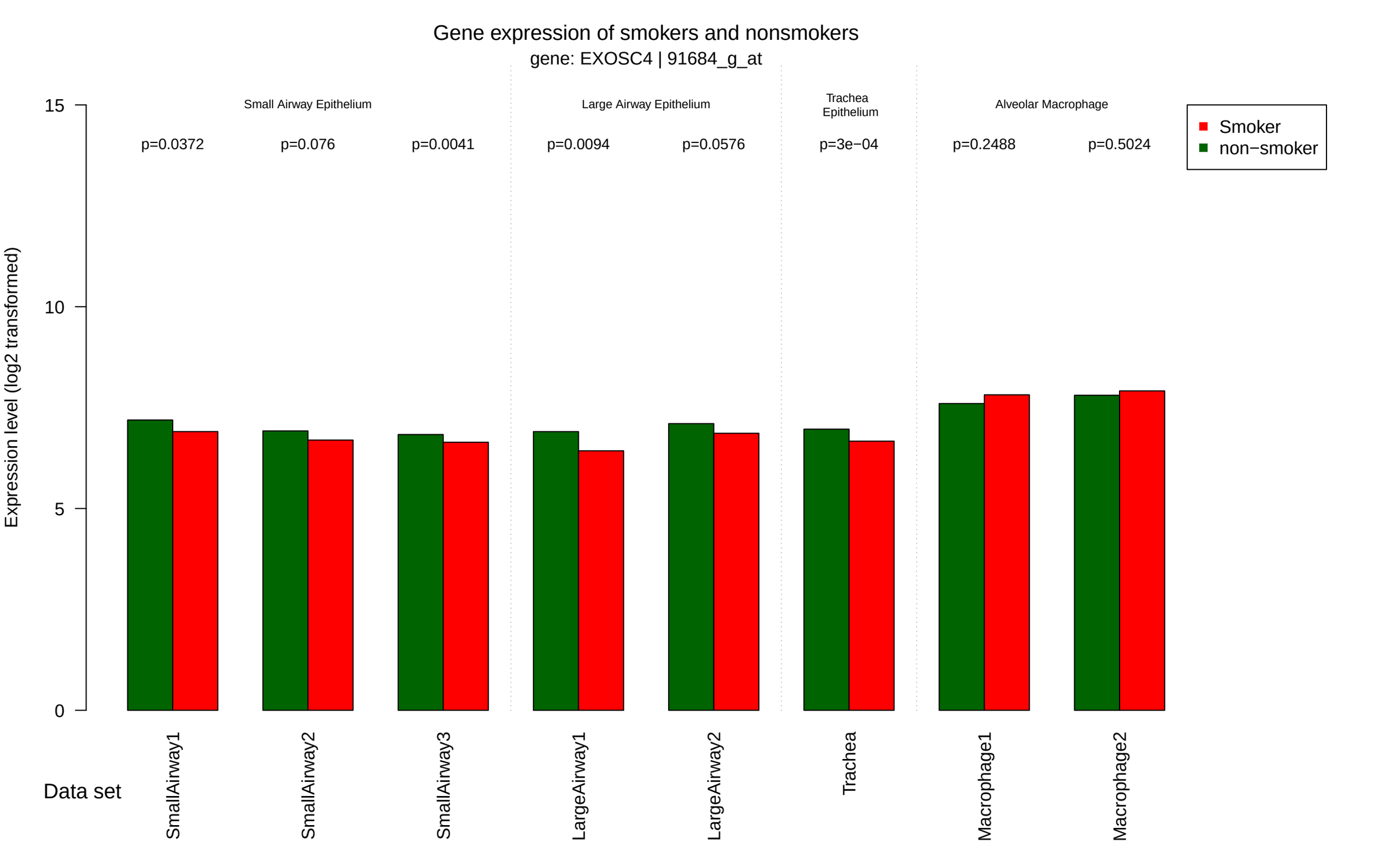
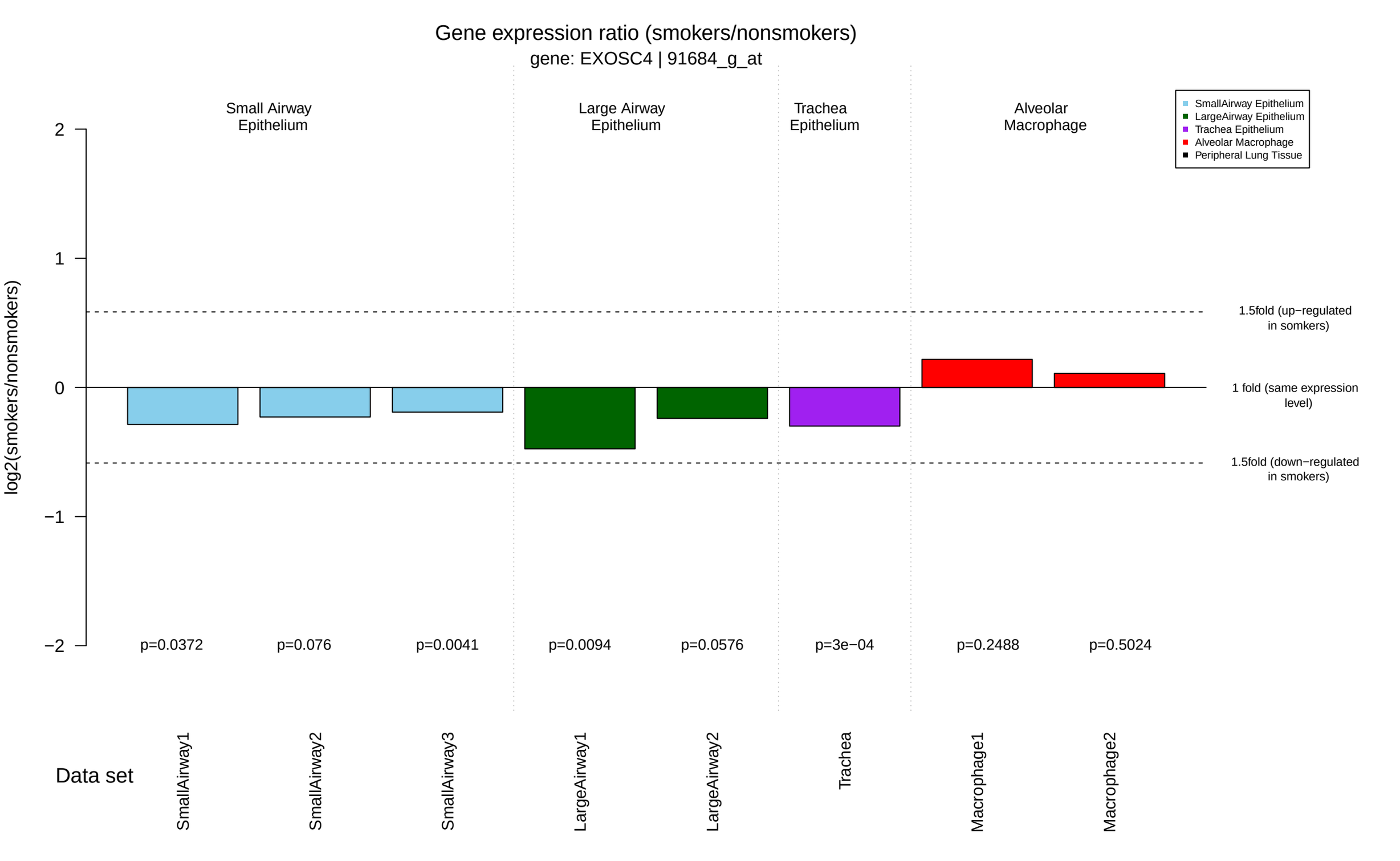
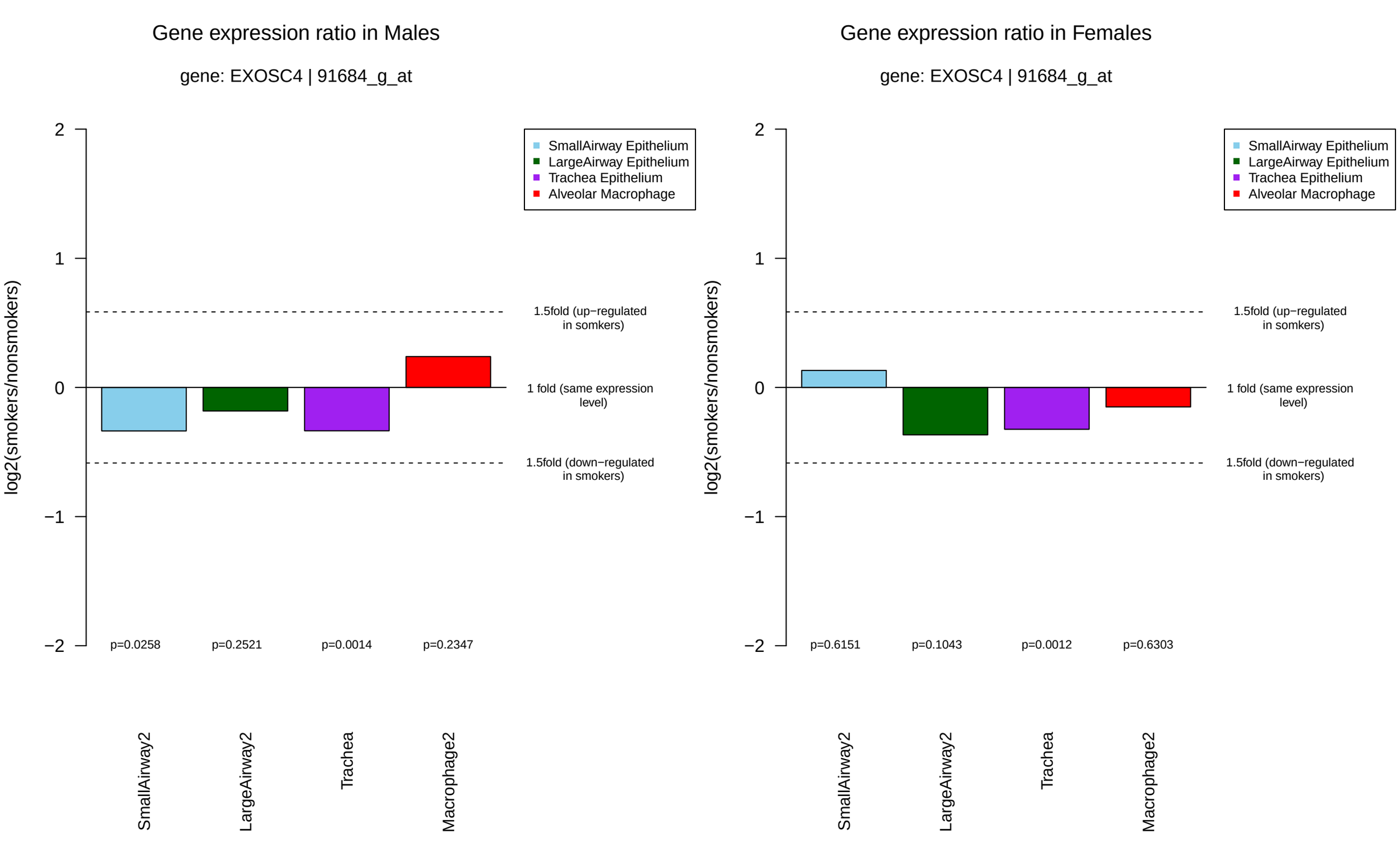
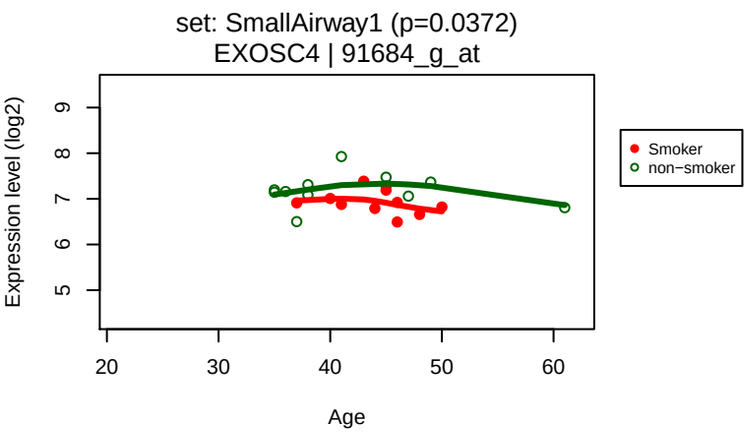
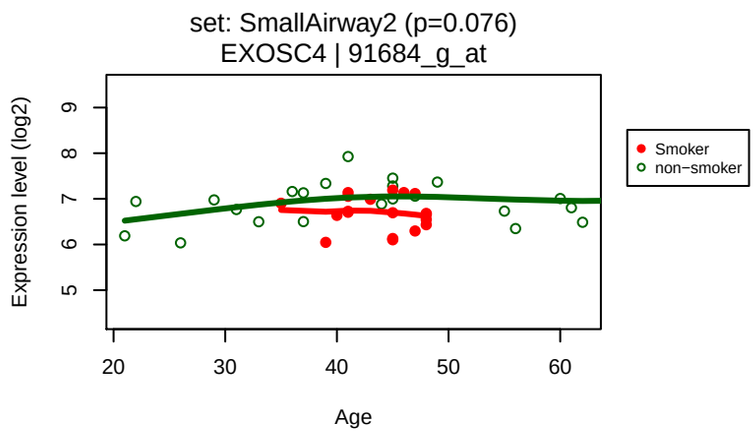
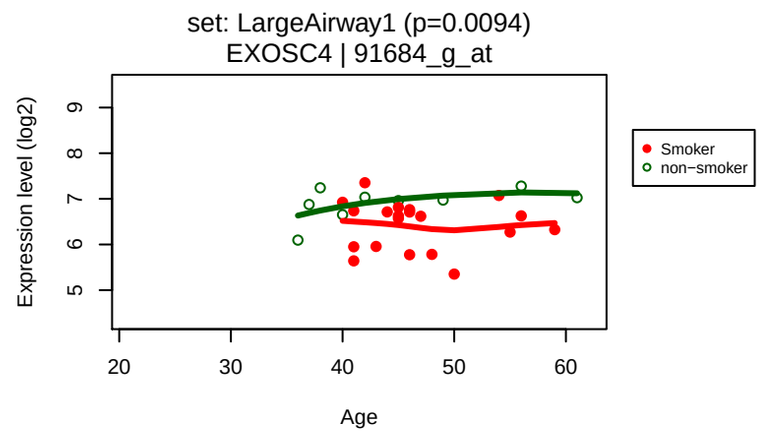

Gene expression of smokers and nonsmokers
gene: EXOSC4 | 91684_g_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0372
p=0.076
p=0.0041
p=0.0094
p=0.0576
p=3e−04
p=0.2488
p=0.5024
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

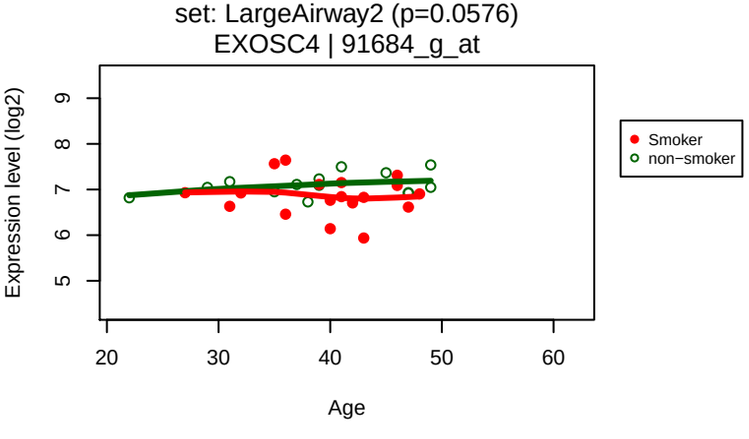

Gene expression ratio (smokers/nonsmokers)
gene: EXOSC4 | 91684_g_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0372
p=0.076
p=0.0041
p=0.0094
p=0.0576
p=3e−04
p=0.2488
p=0.5024
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

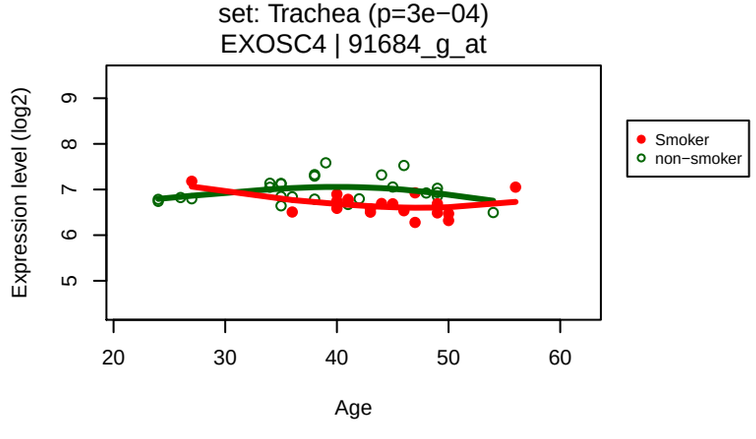

Gene expression ratio in Males
gene: EXOSC4 | 91684_g_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0258
p=0.2521
p=0.0014
p=0.2347
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: EXOSC4 | 91684_g_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6151
p=0.1043
p=0.0012
p=0.6303
SmallAirway2
LargeAirway2
Trachea
Macrophage2

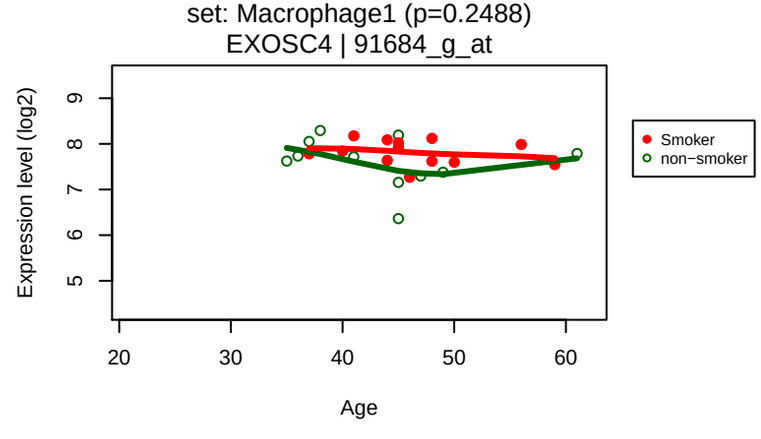

set: SmallAirway1 (p=0.0372)
EXOSC4 | 91684_g_at
Expression level (log2)
Age
Smoker
non−smoker

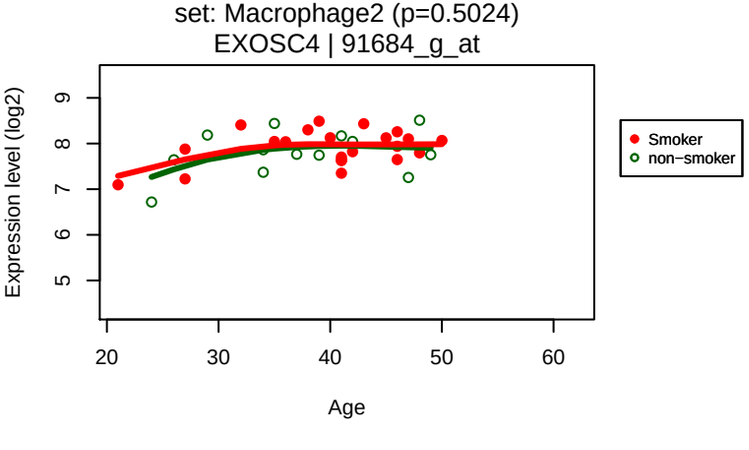

set: SmallAirway2 (p=0.076)
EXOSC4 | 91684_g_at
Expression level (log2)
Age
Smoker
non−smoker

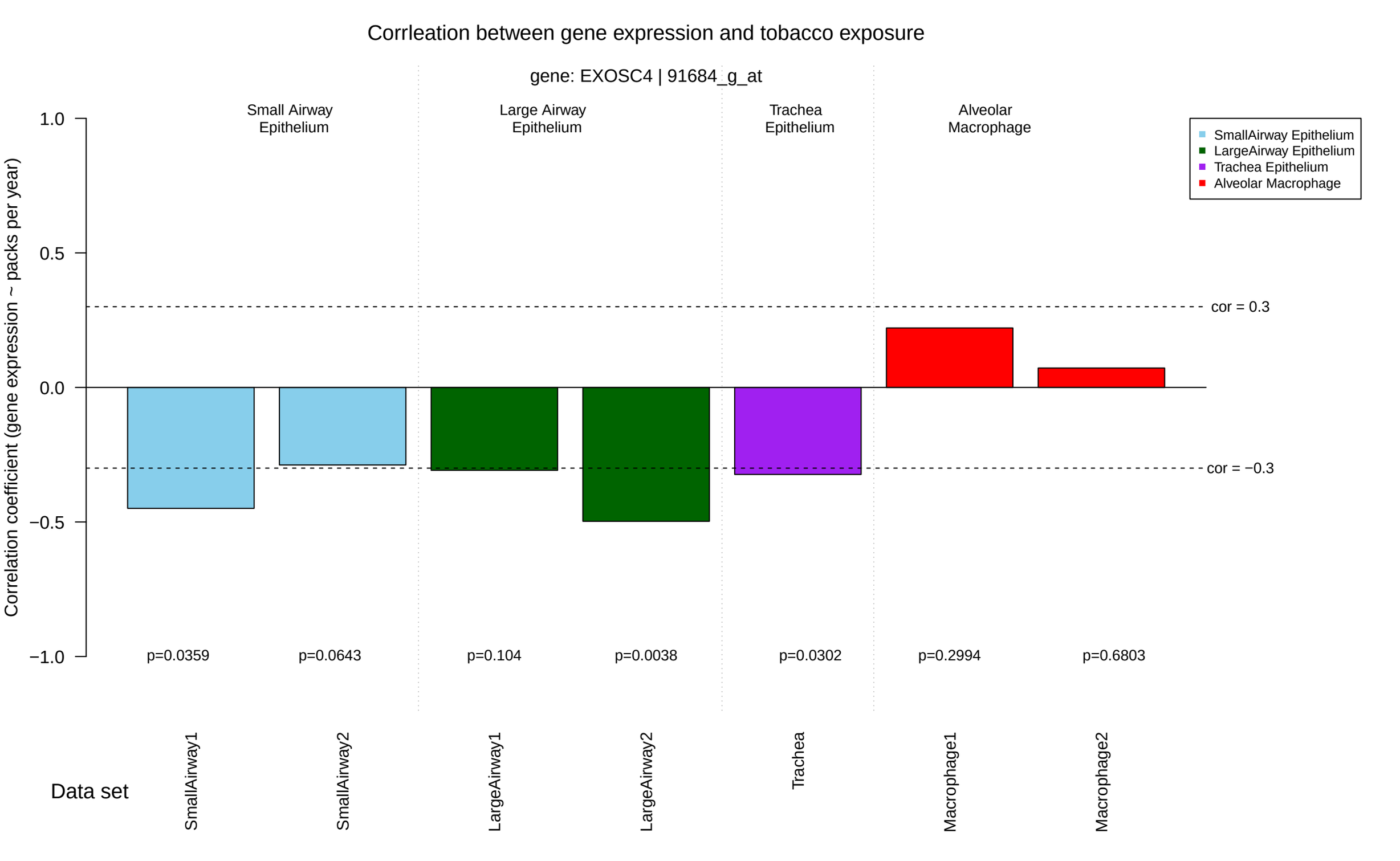

set: LargeAirway1 (p=0.0094)
EXOSC4 | 91684_g_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0576)
EXOSC4 | 91684_g_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=3e−04)
EXOSC4 | 91684_g_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2488)
EXOSC4 | 91684_g_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5024)
EXOSC4 | 91684_g_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: EXOSC4 | 91684_g_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0359
p=0.0643
p=0.104
p=0.0038
p=0.0302
p=0.2994
p=0.6803
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2